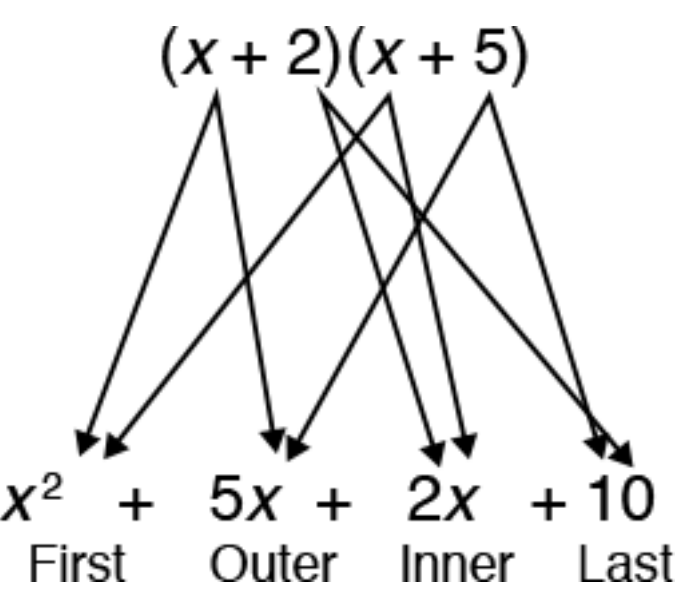


Using FOIL to Factor a Trinomial

You use FOIL to multiply two binomial factors. The answer is a trinomial that looks like this: $ax^2 + bx + c$, where a , b , and c are numbers. You can also use the idea behind FOIL to help you find the binomial factors that make a trinomial.

Look at the diagram below. Can you see how the terms in the binomials are used to find the product?



The simplified trinomial is $x^2 + 7x + 10$. When you add 5 and 2, you get the coefficient of the middle term, $7x$. You multiplied 5 and 2 to get 10, the last term.

When you factor a trinomial, look for a pair of numbers that multiply to make the last term and add to make the middle term.

Example

Hide

Factor: $x^2 + 4x - 12$

Step 1

Find the factors of x^2 .

$x \cdot x = x^2$

Step 2

List the factors of -12 .

1 and -12 -1 and 12

2 and -6 -2 and 6

3 and -4 -3 and 4

Step 3

Look for the factor pair that combines to make 4.

$-2 + 6 = 4$

Step 4

Write the factors.

$(x - 2)(x + 6)$

The factorization of $x^2 + 4x - 12$ is $(x - 2)(x + 6)$.

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Lesson 2: Factoring

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